

Laravel + Docker Complete Guide (Local + Live Server)

1. Introduction

Docker ek containerization platform hai jo aapke application ko uski dependencies ke sath ek isolated environment mai run karta hai.

Is guide mai aap seekhoge:

- Laravel ko Docker mai local run karna
 - Server (live) par deploy karna
 - Basic concepts samajhna
-

2. Docker Kya Hai?

Docker ek system hai jo:

- PHP
- MySQL
- Redis
- Nginx ko alag containers mai run karta hai.

Is se benefit:

- Same environment har machine par
 - No dependency issues
-

3. Local vs Live Server

Local (Development)

- Windows / Laptop
- Docker Desktop use hota hai
- Testing & development ke liye
- URL: `http://localhost:8000`

Live Server (Production)

- Linux VPS (Ubuntu)
- Docker Engine install hota hai
- Real users access karte hain
- URL: `https://yourdomain.com`

4. Required Tools

Local:

- Docker Desktop
- Laravel Project
- Code Editor

Live Server:

- Ubuntu Server
 - SSH access
 - Docker Engine
 - Docker Compose
-

5. Laravel Docker Structure

Project structure:

```
my-laravel-app/  
|  
├─ Dockerfile  
├─ docker-compose.yml  
├─ .env  
├─ nginx/  
│   └─ default.conf  
├─ app/  
└─ artisan
```

6. Dockerfile

```
FROM php:8.2-fpm  
  
WORKDIR /var/www  
  
RUN apt-get update && apt-get install -y  
    zip unzip curl git libpng-dev  
  
RUN docker-php-ext-install pdo pdo_mysql  
  
COPY . .
```

```
COPY --from=composer:latest /usr/bin/composer /usr/bin/composer
```

```
CMD ["php-fpm"]
```

7. docker-compose.yml

```
version: '3.8'

services:

  app:
    build: .
    container_name: laravel_app
    working_dir: /var/www
    volumes:
      - ./:/var/www
    depends_on:
      - mysql

  nginx:
    image: nginx:latest
    container_name: nginx_server
    ports:
      - "8000:80"
    volumes:
      - ./:/var/www
      - ./nginx/default.conf:/etc/nginx/conf.d/default.conf
    depends_on:
      - app

  mysql:
    image: mysql:8
    container_name: mysql_db
    restart: always
    environment:
      MYSQL_DATABASE: laravel
      MYSQL_ROOT_PASSWORD: root
    ports:
      - "3307:3306"
```

8. Nginx Config

```
server {
    listen 80;
```

```
index index.php;
root /var/www/public;

location / {
    try_files $uri $uri/ /index.php?$query_string;
}

location ~ \.php$ {
    fastcgi_pass app:9000;
    fastcgi_index index.php;
    fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
    include fastcgi_params;
}
}
```

9. .env Configuration

```
DB_CONNECTION=mysql
DB_HOST=mysql
DB_PORT=3306
DB_DATABASE=laravel
DB_USERNAME=root
DB_PASSWORD=root
```

10. Local Setup Steps

Step 1: Start containers

```
docker compose up -d --build
```

Step 2: Enter container

```
docker exec -it laravel_app bash
```

Step 3: Install dependencies

```
composer install
```

Step 4: Setup Laravel

```
php artisan key:generate  
php artisan migrate
```

Step 5: Open browser

```
http://localhost:8000
```

11. Live Server Setup Steps

Step 1: Server login

```
ssh root@server-ip
```

Step 2: Install Docker

```
apt update  
apt install docker.io -y  
apt install docker-compose -y
```

Step 3: Clone project

```
git clone your-repo  
cd your-project
```

Step 4: Run project

```
docker compose up -d --build
```

Step 5: Access site

```
http://server-ip
```

or domain connect

12. Key Concepts Summary

Docker does:

- App ko isolate karta hai
- Same environment har jagah
- Dependency issues khatam

You still need:

- Docker install (local + server)
 - docker-compose run command
 - Laravel setup commands (first time)
-

13. Important Notes

✓ First time composer install zaroor hota hai ✓ DB_HOST container name hota hai (mysql) ✓ Same docker setup local aur live mai use hota hai ✓ Production mai SSL + domain add hota hai

14. Final Flow

Local:

```
docker compose up -d --build
composer install
php artisan migrate
```

Live:

```
git clone repo
docker compose up -d --build
```

END

Agar aap chaho to next step mai mai aapko:

- Production-ready Docker setup (queues + redis + supervisor)
- CI/CD auto deploy (GitHub Actions)

bhi sikha sakta hoon.